

```

#include "glut.h"
#include <cmath>
using namespace std;

void drawRect(float x, float y, float width, float height, float r, float g, float b) {
    glColor3f(r, g, b);
    glBegin(GL_QUADS);
    glVertex2f(x, y);
    glVertex2f(x + width, y);
    glVertex2f(x + width, y + height);
    glVertex2f(x, y + height);
    glEnd();
}

void drawCircle(float cx, float cy, float radius, float r, float g, float b) {
    glColor3f(r, g, b);
    glBegin(GL_TRIANGLE_FAN);
    glVertex2f(cx, cy);
    for (int i = 0; i <= 100; i++) {
        float angle = 2.0f * 3.14159f * i / 100;
        glVertex2f(cx + radius * cos(angle), cy + radius * sin(angle));
    }
    glEnd();
}

void drawTriangle(float x, float y, float size, float r, float g, float b) {
    glColor3f(r, g, b);
    glBegin(GL_TRIANGLES);
    glVertex2f(x, y);
    glVertex2f(x + size, y);
    glVertex2f(x + size / 2, y + size);
    glEnd();
}

void display() {
    glClear(GL_COLOR_BUFFER_BIT);

    glLineWidth(2.0);

    glColor3f(1.0f, 0.0f, 0.0f);
    glBegin(GL_LINES);
    glVertex2f(-1.0f, 0.0f);
    glVertex2f(1.0f, 0.0f);
    glEnd();

    glColor3f(0.0f, 1.0f, 0.0f);

```

```
glBegin(GL_LINES);
glVertex2f(0.0f, -1.0f);
glVertex2f(0.0f, 1.0f);
glEnd();
```

```
drawRect(-0.6f, -0.6f, 1.2f, 1.2f, 0.5f, 0.2f, 0.2f);
drawRect(-0.6f, -0.6f, 0.05f, 1.2f, 0.3f, 0.1f, 0.1f);
drawRect(0.55f, -0.6f, 0.05f, 1.2f, 0.3f, 0.1f, 0.1f);
drawRect(-0.55f, -0.6f, 1.1f, 0.05f, 0.3f, 0.1f, 0.1f);
drawRect(-0.55f, 0.55f, 1.1f, 0.05f, 0.3f, 0.1f, 0.1f);
```

```
drawRect(-0.55f, -0.3f, 1.1f, 0.05f, 0.3f, 0.1f, 0.1f);
drawRect(-0.55f, 0.0f, 1.1f, 0.05f, 0.3f, 0.1f, 0.1f);
drawRect(-0.55f, 0.3f, 1.1f, 0.05f, 0.3f, 0.1f, 0.1f);
```

```
drawCircle(-0.45f, 0.42f, 0.05f, 0.4f, 1.0f, 1.0f);
```

```
drawTriangle(-0.3f, 0.37f, 0.1f, 1.0f, 0.3f, 1.0f);
```

```
drawRect(-0.5f, -0.25f, 0.03f, 0.25f, 0.3f, 1.0f, 0.7f);
drawRect(-0.45f, -0.25f, 0.03f, 0.25f, 0.3f, 1.0f, 0.7f);
drawRect(-0.4f, -0.25f, 0.03f, 0.25f, 0.3f, 1.0f, 0.7f);
```

```
glPushMatrix();
glTranslatef(-0.52f, -0.58f, 0.0f);
glRotatef(-45, 0.0f, 0.0f, 1.0f);
drawRect(0.0f, 0.0f, 0.03f, 0.2f, 0.3f, 1.0f, 1.0f);
glPopMatrix();
```

```
glPushMatrix();
glTranslatef(0.4f, 0.05f, 0.0f);
glRotatef(30, 0.0f, 0.0f, 1.0f);
drawRect(0.0f, 0.0f, 0.03f, 0.2f, 0.5f, 1.0f, 0.5f);
glPopMatrix();
```

```
glFlush();
```

```
}
```

```
void init() {
    glClearColor(1.0, 1.0, 1.0, 1.0);
    glMatrixMode(GL_PROJECTION);
    gluOrtho2D(-1.0, 1.0, -1.0, 1.0);
```

```
}  
  
int main(int argc, char** argv) {  
    glutInit(&argc, argv);  
    glutInitWindowSize(800, 800);  
    glutCreateWindow("Gia Sach - OpenGL");  
    init();  
    glutDisplayFunc(display);  
    glutMainLoop();  
    return 0;  
}
```

